```c
void produce(int item) {
    if (count == SIZE) {
        printf("Buffer FULL!\n");
        return;
    }

    buffer[in] = item;
    in = (in + 1) % SIZE;
    count++;

    printf("Produced: %d\n", item);
}

void consume() {
    if (count == 0) {
        printf("Buffer EMPTY!\n");
        return;
    }

    int item = buffer[out];
    out = (out + 1) % SIZE;
    count--;

    printf("Consumed: %d\n", item);
}

int main() {
    int choice, item;

    while (1) {
        printf("\n1. Produce\n2. Consume\n3. Exit\n");
        printf("Enter choice: ");
        scanf("%d", &choice);

        switch (choice) {
            case 1:
                printf("Enter item: ");
                scanf("%d", &item);
                produce(item);
                break;

            case 2:
                consume();
                break;

            case 3:
                return 0;

            default:
                printf("Invalid choice!\n");
        }
    }

    return 0;
}
"producer_consumer.c" 62L, 1068B                                    62,1          Bot
```

Apr 30 1:23 PM

ashvanthanachu@archlinux:~

```
[ashvanthanachu@archlinux ~]$ gcc producer_consumer.c -o p_c.out
[ashvanthanachu@archlinux ~]$ ./p_c.out

1. Produce
2. Consume
3. Exit
Enter choice: 1
Enter item: 2
Produced: 2

1. Produce
2. Consume
3. Exit
Enter choice: 1
Enter item: 1
Produced: 1

1. Produce
2. Consume
3. Exit
Enter choice: 1
Enter item: 2
Produced: 2

1. Produce
2. Consume
3. Exit
Enter choice: 1
Enter item: 2
Produced: 2

1. Produce
2. Consume
3. Exit
Enter choice: 1
Enter item: 2
Produced: 2

1. Produce
2. Consume
3. Exit
Enter choice: 2
Consumed: 2

1. Produce
2. Consume
3. Exit
Enter choice: 2
Consumed: 1

1. Produce
2. Consume
3. Exit
Enter choice: 3
[ashvanthanachu@archlinux ~]$ vim producer_consumer.c
[ashvanthanachu@archlinux ~]$
```